

Comp 442 Group Assignment 4
“Phish Detector”

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Project Overview

Project Goal

- Develop a machine learning-based spam and phishing detector
- Use NLP and supervised learning

Assignment Objective

- Prepare the dataset for future machine learning
- Improve data quality
- Generate useful features

Dataset Overview

Dataset Information

- Email Spam Detection Dataset
- 5,572 records
- 2 attributes
 - Category
 - Message

Initial Data Quality

- No missing values
- 415 duplicate records
- Outliers identified
- Class imbalance

Data Cleaning

Missing Values

- None detected

Duplicate Detection

- 415 duplicates

Duplicate Removal

- Dataset reduced to 5,157 records

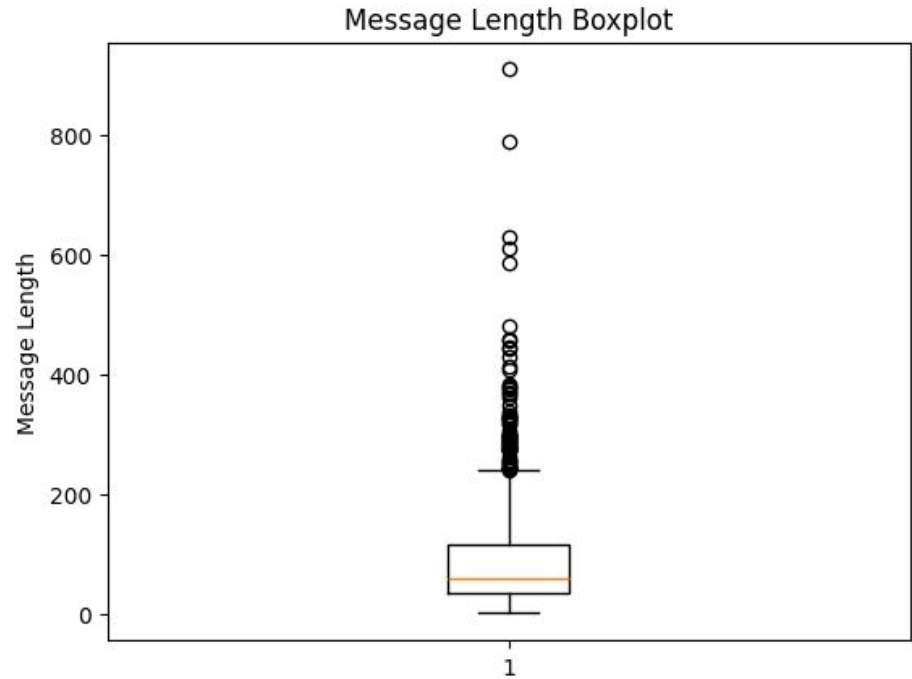
Image

- Duplicate removal output from Colab

Outlier Detection

Key Findings

- Several long-message outliers
- Positive skewness
- Outliers retained



Data Transformation

Discretization

- Short
- Medium
- Long

Normalization

- Min-Max Scaling

Images

- Length Category output
- Normalized Message Length output

Feature Selection

Selected Features

- Message Length
- Word Count
- Digit Count
- URL Count
- Length Category
- Normalized Message Length

Why?

- Improve model performance
- Reduce complexity
- Focus on informative features

Feature Extraction

Created Features

- Message Length
- Word Count
- Digit Count
- URL Count

Purpose

- Convert raw text into measurable numerical attributes

Results & Findings

Data Cleaning

- No missing values
- Duplicates removed

Transformation

- Discretization completed
- Normalization completed

Feature Engineering

- Six useful features prepared

Conclusion

Summary

- Dataset successfully cleaned
- Data transformed into model-ready format
- Features extracted for classification

Next Steps

- TF-IDF Vectorization
- Model Training
- Model Evaluation
- Deployment